

Rosenblatt

$$E[I_n^{(1)}] = (nb^2)^{-1} \left(\int \hat{w}(x_i) dx_i \right)^2$$

$$\text{Var}[I_n^{(1)}] = O_p((nb)^{-1})$$

$$I_n^{(2)} = \sqrt{2} \sqrt{(nb)^{-1}} N_n$$

$$E[I_n] = (nb)^{-1} L + o((nb)^{-1})$$

$$E[I_{n3}] = -E[I_{n2}]$$

$$\text{Var}(I_{n2}) = O((nb)^{-1})$$

$$\text{Var}(I_{n3}) = \text{Var}(I_{n2})$$

Me

$$E[I_n] = \frac{\sqrt{d(L)}}{n\rho} (1 + o(1))$$

$$\text{Var}[I_{n2}] = O_p((n^{\frac{3}{2}}\rho)^{-1})$$

$$I_{n,3} \approx \sqrt{2} \lambda^{\frac{1}{2}} \sqrt{\rho} n^{-1} \rho^{-1/2} N_n$$

$$E[T_{n,2}] = \sum_{j=1}^n \frac{v_j R_j}{n h_j^{\alpha_j}} (1 + o(1))$$

$$E[T_{n,3}] = -2E[T_{n,2}]$$

$$\text{Var}(T_{n,2}) = O(n^{-1} \sum_{j=1}^n h_j^{-\alpha_j})$$

$$\text{Var}(T_{n,3}) = \text{Var}(T_{n,2})$$

CASE: $n=2$, $d_1=d_2=1$, $h_1=h_2=b$. ($b=\sqrt{p}$) ($p=b^2$)

Rosenblatt

$$E[I_{n,1}^{(1)}] = (nb^2)^{-1} \left(\int w^2(x_1) dx_1 \right)^2 \asymp$$

$$\text{Var}[I_{n,1}^{(1)}] = o_p((nb)^{-1}) \quad *$$

$$I_{n,1}^{(2)} = \sqrt{2} v (nb)^{-1} N_n \asymp$$

$$E[I_{n,1}] = (nb)^{-1} L + o((nb)^{-1}) \quad \asymp$$

$L < \text{CONSTANT}$

$$E[I_{n,3}] = -E[I_{n,2}] \asymp$$

$$\text{Var}(I_{n,1}) = o((n^2 b)^{-1}) \asymp$$

$$\text{Var}(I_{n,3}) = \text{Var}(I_{n,1}) \asymp$$

Me

$$E[I_{n,1}] = \frac{\sqrt{d(1)}}{n b^2} (1 + o(1))$$

$$\text{Var}[I_{n,1}] = O_p((n^2 b^2)^{-1})$$

$$I_{n,3} \asymp \sqrt{2} \lambda \sqrt{p} n^{-1} b^{-1} N_n$$

$$E[T_{n,2}] = \frac{v_1 R_1 + v_2 R_2}{n b} (1 + o(1))$$

$$E[T_{n,3}] = -2E[T_{n,2}]$$

$$\text{Var}(T_{n,1}) = O(n^{-2} b^{-1})$$

$$\text{Var}(T_{n,3}) = \text{Var}(T_{n,2})$$

*: MY ORDER IS EVEN BETTER